THE BINVENTORS

BINARDO BINSTEIN

AI-POWERED SMART WASTE SORTER



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THE PROBLEM

India throws away 62 million tonnes a year

Waste is not a small problem in India. It is a national one — and almost all of it is being buried instead of recovered.

62

MILLION TONNES OF WASTE GENERATED IN INDIA EVERY YEAR

30%

IS ALL THAT GETS RECYCLED

Wrong sorting makes it much worse.

Source: Ministry of Environment, Forest and Climate Change (MoEFCC); Earth5r

WHY IT FAILS

03

One wrong item ruins the whole bin

Most people have no idea which bin to use. That is where recycling breaks.



CONTAMINATION

When wet food mixes with paper or plastic, the whole bin becomes useless. Even recyclable material ends up in a landfill.



LANDFILL

Less than 30% of India's waste is recycled. Every mis-sorted bin pushes that number lower.

1 WRONG ITEM = THE ENTIRE BIN BECOMES UNRECYCLABLE

OUR EXPERIENCE

We watched it happen at our own school

Watching our school janitors manually sort through piles of sticky, mixed trash was a wake-up call. It exposed the brutal reality of waste segregation confusion on campus,and forced human workers to do the dirty work. This everyday mess is a systemic failure that undermines India's Swachh Bharat mission.

Someone is already doing this job. By hand.





04

OUR SURVEY · 92 RESPONDENTS · AUG 2026

05

We asked 92 people what they actually want

86%

WANT AN AI-SORTING BIN

85.9% said an auto-sorting bin would be “very useful”.

47%

WANT CLEARER BIN LABELS

47.3% want clearer labels, the top-requested fix.

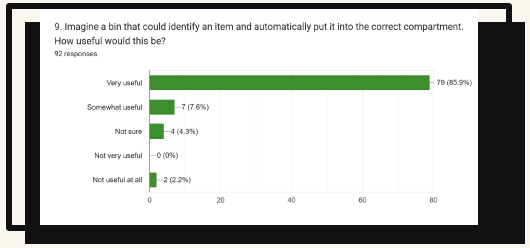
43%

WANT THE BIN TO DECIDE FOR THEM

42.9% want a system that decides for them.

People don't want better instructions. They want to stop deciding.





Raw responses from our survey form.

Source: Our Waste Segregation Survey, 92 respondents (Aug 2026)

OUR SOLUTION

06

Meet Binardo Binstein



The working prototype.

Binardo Binstein uses camera detection to identify waste and automatically sort it into the correct compartment — so nobody has to guess.

SORTS AUTOMATICALLY

Drop in waste — Binardo decides which compartment it goes to.

AI-POWERED

The AI identifies the waste type every single time.

TEACHES

Helps students learn good sorting habits.

ROBUST

Built from aluminium rods for daily school use.

HOW IT WORKS

Detection, decision, sorting

Binardo does all the hard work — here's how.

1

DETECTION

The camera scans the waste and identifies what type it is.

2

DECISION

The system decides which category the waste belongs to: paper, food, plastic or metal.

3

SORTING

The waste is automatically directed into the correct compartment.



VIDEO DEMONSTRATION

Watch Binardo sort a real item, live.

COSTING

08

From Rs. 44,000 to under Rs. 25,000

Every rupee we cut brings Binardo Binstein closer to every classroom.

CURRENT PROTOTYPE

Rs. 44,000

(about $461) per unit

Built around a Raspberry Pi for camera detection and sorting logic. It works well, but the Pi is the single most expensive part.

43%
COST CUT

OUR PLAN

UNDER

Rs. 25,000

(about $262) per unit

Swap the Raspberry Pi for an ESP32 microcontroller, the same detection logic at fraction of the cost, with some loss of accuracy.

Cheap enough for a school to buy one out of its own budget.

Source: Retail pricing for Raspberry Pi 4 and ESP32 boards (Aug 2026)

UN SDGS WE SOLVE

Three UN Global Goals



GOAL 13 CLIMATE ACTION

Take urgent action to combat climate change and its impacts.



GOAL 12 RESPONSIBLE USE

Ensure sustainable consumption and production patterns.



GOAL 11 SUSTAINABLE CITIES

Make cities and human settlements inclusive, safe, resilient and sustainable.

Following these goals means helping our planet.

FUTURE IMPACT

What changes when waste is sorted properly

1

SAFEGUARDING HEALTH

It improves public health by reducing the breeding grounds for pests that spread disease, and cuts human exposure to toxic substances.

2

HIGHER RECOVERY

AI sorting significantly boosts recycling yields across plastic, paper, cloth and other materials.

3

PROTECTING THE ECOSYSTEM

Handling waste removal scientifically minimises the air, soil and water pollution caused by open burning and dumping.

Source: Northamps; ReKart

SWOT ANALYSIS

What we're strong at — and what we're not

We built it, so we know what it can't do yet.

S · STRENGTHS

- AI-powered sorting at the source
- 86% of surveyed users want automatic sorting

W · WEAKNESSES

- The AI must be retrained for new materials
- Higher cost than a conventional bin
- Segregates one waste item at a time

O · OPPORTUNITIES

- Build a conveyor-based industrial sorter
- Deploy across schools, offices and other sites
- Partner with recyclers such as Jaagruti waste paper recycling services

T · THREATS

- Conventional bins are far cheaper, so customers may question paying more for an intelligent bin
- Real-world waste is messy — wet, crushed, dirty or partially concealed items could reduce AI accuracy

Video Presentation and Live Demo of the Binardo Binstein

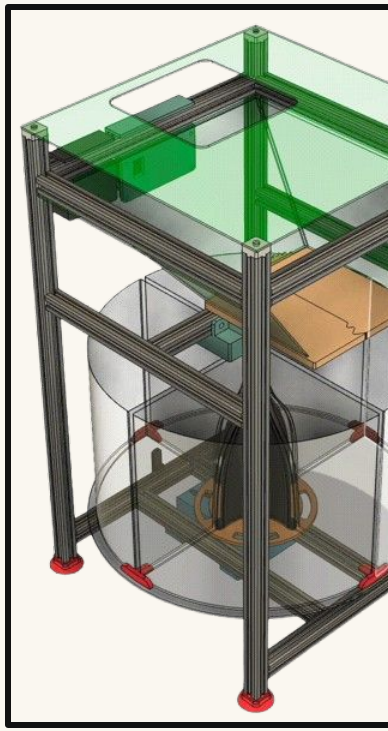


Video Link : https://www.youtube.com/watch?v=vnWBGc1xFEA

WE DIDN'T ADD A BIN. WE REMOVED A DECISION.

Every other fix asks people to sort better. Binardo makes sorting something they no longer have to do.

Thank you. Questions? We'd love to hear them.



SOURCES

Our Waste Segregation Survey (92 respondents, Aug 2026) Ministry of Environment, Forest and Climate Change (MoEFCC) Earth5r Jaagruti waste paper recycling services

UN Sustainable Development Goals Retail pricing, Raspberry Pi 4 and ESP32 (Aug 2026) Northamps ReKart